

XITAI, China

WMO: 588530

Lat: 27.008N

Lon: 120.695E

Elev: 351

StdP: 14.51

Time Zone: 8.00 (E08)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	37.5	38.9	19.1	14.6	42.0	22.9	17.5	42.3	40.9	42.2	38.4	43.0	24.3	20	0.716

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	85.9	79.6	84.5	79.3	83.3	78.8	80.6	83.9	80.0	82.9	79.4	82.0	10.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.7	156.0	82.4	79.1	152.9	81.8	78.6	150.2	81.2	44.6	84.6	44.0	83.3	43.4	82.5	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.7	36.8	33.7	DB	34.7	89.3	3.0	2.7	32.5	91.3	30.7	92.9	29.0	94.5	26.8	96.5	(4)
(5)			WB	29.9	82.0	2.2	0.6	28.3	82.5	26.9	82.9	25.7	83.2	24.1	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	47.3	46.5	49.9	57.5	66.7	73.9	78.9	80.7	76.9	69.3	61.6	51.9		
	DBStd	12.87	4.70	4.69	4.57	4.80	4.49	3.48	2.57	2.11	3.75	4.19	4.76	5.85		
	HDD50	340	109	118	58	4	0	0	0	0	0	0	2	49		
	HDD65	2331	548	518	468	228	36	0	0	0	0	10	118	405		
	CDD50	5279	27	19	55	230	518	717	897	950	807	599	351	109		
	CDD65	1796	0	0	0	4	88	268	432	485	357	145	17	0		
	CDH74	11757	0	0	0	0	45	1030	3530	4586	2336	226	4	0		
	CDH80	1579	0	0	0	0	1	39	410	847	267	15	0	0		
(14) Wind	WSAvg	18.2	20.2	20.1	18.1	14.4	14.9	16.8	18.9	15.7	18.7	20.7	21.0	19.2		
(15) Precipitation	PrecAvg	43.3	2.0	2.9	4.7	5.0	6.1	7.1	1.8	3.9	4.6	2.0	1.9	1.3		
	PrecMax	58.9	6.7	5.6	11.6	12.9	16.3	15.4	6.0	11.9	13.2	7.6	6.1	3.7		
	PrecMin	18.3	0.1	0.6	0.7	0.3	1.1	1.2	0.2	0.3	0.3	0.0	0.0	0.0		
	PrecStd	9.9	1.7	1.3	3.0	2.9	3.5	3.8	1.8	2.6	3.4	1.8	1.5	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	61.7	64.4	71.0	77.4	82.7	86.8	88.6	86.0	82.1	73.1	65.6		
		MCWB	55.3	56.7	61.7	67.7	72.3	78.8	80.6	79.5	79.6	77.1	68.8	59.5		
	2%	DB	57.6	58.1	60.7	67.9	75.3	81.0	84.9	86.6	84.3	78.2	70.1	62.5		
		MCWB	53.6	55.3	58.4	65.8	72.1	77.9	80.0	79.4	78.8	75.1	66.9	58.2		
	5%	DB	55.4	55.4	58.5	66.1	73.8	79.8	83.7	85.2	82.7	76.2	68.8	60.9		
		MCWB	52.5	52.4	56.7	64.5	71.5	77.3	79.4	79.2	78.1	73.1	65.9	56.9		
	10%	DB	53.4	52.8	56.5	64.3	72.4	78.6	82.4	84.0	81.5	74.5	67.5	59.4		
		MCWB	50.7	50.4	54.4	62.0	70.6	76.6	78.9	79.0	77.6	70.5	64.6	55.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.2	58.9	62.6	68.7	74.6	79.7	81.7	81.5	80.7	77.9	69.7	61.5		
		MCDB	59.4	60.2	63.8	70.2	75.9	81.5	85.1	85.2	83.7	80.6	71.2	63.6		
	2%	WB	54.9	56.0	59.4	66.5	73.4	78.5	80.6	80.7	79.8	76.3	68.2	59.7		
		MCDB	56.7	57.6	60.1	67.2	74.4	80.0	83.5	84.4	82.5	77.8	69.6	61.9		
	5%	WB	53.0	53.1	57.5	64.8	72.2	77.8	79.9	80.0	79.0	74.6	67.0	58.0		
		MCDB	54.9	54.5	58.4	65.9	73.2	79.2	82.5	83.4	81.7	75.9	68.4	60.3		
	10%	WB	51.0	51.0	54.8	62.9	71.0	77.0	79.2	79.5	78.3	71.8	65.7	56.0		
		MCDB	53.0	52.5	56.1	64.0	72.0	78.3	81.5	82.5	80.8	73.8	67.2	59.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	4.7	5.0	5.8	5.4	4.8	5.2	6.0	5.1	4.5	4.8	5.4		
		MCDBR	9.3	9.4	8.4	8.3	7.5	5.9	7.2	8.0	6.7	6.4	6.5	7.6		
	5% WB	MCWBR	6.5	7.1	6.9	6.8	5.1	3.2	3.6	3.6	3.8	4.4	5.3	6.2		
		MCWBR	8.4	7.4	8.5	7.9	6.3	5.2	6.5	7.0	6.1	6.2	6.3	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.575	0.625	0.684	0.685	0.620	0.580	0.537	0.565	0.556	0.561	0.559	0.547		
		taud	1.754	1.689	1.602	1.660	1.841	1.971	2.101	2.012	2.001	1.899	1.860	1.829		
	Ebn at Noon		206	209	207	213	227	235	245	238	235	223	210	207		
		Edh at Noon	63	72	83	80	67	58	51	56	55	57	56	56		
(44) All-Sky Solar Radiation	RadAvg		689	784	1039	1194	1323	1425	1945	1742	1426	1173	786	720		
	RadStd		155	180	107	144	205	164	154	142	140	196	119	132		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page